

Solution

4. (Cake division). Albert and Bernie divide a cake by alternating proposals. First Albert proposes a division. If Bernie accepts, the division is implemented. Otherwise they wait one hour and Bernie proposes. If Albert accepts, the division is implemented; otherwise they wait another hour, and so on. A fraction p of the cake is worth

$$p \left(\frac{1}{2} \right)^t$$

after t hours of waiting, for $t \leq 5$. After 5 hours the cake becomes worthless. Find a subgame perfect Nash equilibrium and determine how long they wait.

Solution.

Let

$$\delta = \frac{1}{2}.$$

The last possible valuable proposal is at $t = 5$, made by Bernie. If Albert rejects at $t = 5$, both get 0. Hence Bernie can keep the whole remaining cake:

$$V_5^P = \delta^5 = \frac{1}{32}.$$

Work backwards. At time t , the responder can reject and become proposer at time $t + 1$. Therefore the responder must receive exactly the continuation value

$$V_{t+1}^P.$$

Hence

$$V_t^P = \delta^t - V_{t+1}^P.$$

Now compute:

$$V_5^P = \frac{1}{32},$$

$$V_4^P = \frac{1}{16} - \frac{1}{32} = \frac{1}{32},$$

$$V_3^P = \frac{1}{8} - \frac{1}{32} = \frac{3}{32},$$

$$V_2^P = \frac{1}{4} - \frac{3}{32} = \frac{5}{32},$$

$$V_1^P = \frac{1}{2} - \frac{5}{32} = \frac{11}{32},$$

$$V_0^P = 1 - \frac{11}{32} = \frac{21}{32}.$$

At $t = 0$, Albert is proposer. He gives Bernie her continuation value:

$$\frac{11}{32}.$$

Therefore Albert proposes

$$\left(\frac{21}{32}, \frac{11}{32}\right),$$

and Bernie accepts.

Thus the subgame perfect Nash equilibrium outcome is

$$\boxed{\left(\frac{21}{32}, \frac{11}{32}\right)}$$

and they wait

$$\boxed{0 \text{ hours.}}$$